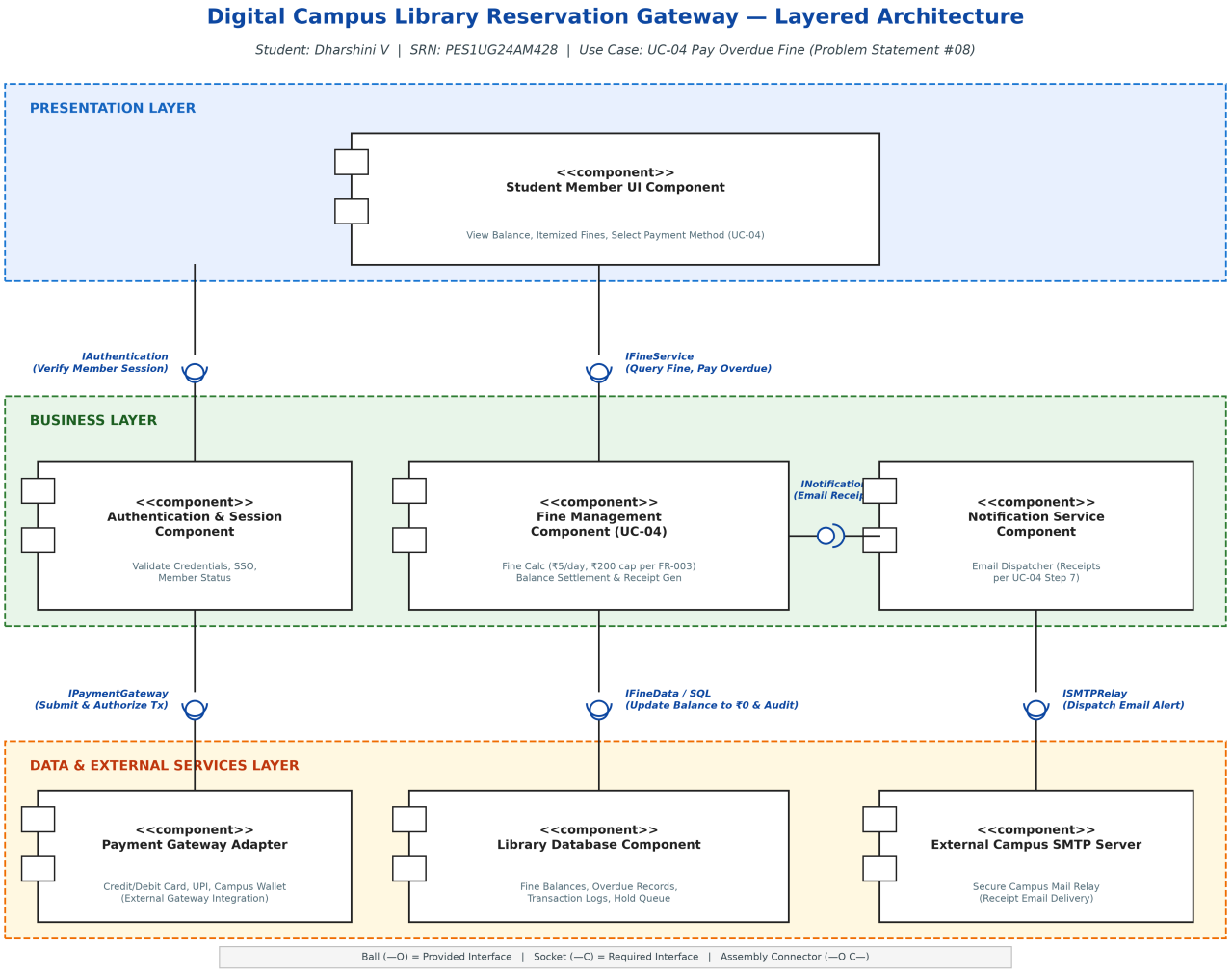
LAB 3 – COMPONENT MODELLING & ARCHITECTURE

Digital Campus Library Reservation Gateway — Problem Statement #08

Student: Dharshini V	**SRN:** PES1UG24AM428
Problem Statement: #08 Library Reservation	**Assigned Use Case:** UC-04 Pay Overdue Fine

Component Diagram

Figure 1. Digital Campus Library Reservation Gateway – Layered Architecture UML Component Diagram



Architecture Selection & Technical Justification

Architecture Selection: We chose **Layered Architecture** for the Digital Campus Library Reservation Gateway.

Architectural Choice

A three-layer architectural style is implemented: **Presentation Layer** (Student Member UI Component), **Business Layer** (Authentication & Session Component, Fine Management Component for UC-04, and Notification Service Component), and **Data & External Services Layer** (Payment Gateway Adapter, Library Database Component, and External Campus SMTP Server). Communication strictly traverses adjacent layers through standardized ball-and-socket assembly interfaces (*IAuthentication, IFineService, IPaymentGateway, IFineData/SQL, INotification, ISMTPRelay*).

Reason 1: Separation of Concerns

The functional requirement for UC-04 (Pay Overdue Fine) and FR-003 naturally separates into distinct operational responsibilities: customer interaction and fine presentation, business calculation rules (computing ■5/day overdue capped at ■200), and financial transaction authorization. Separating these operations into independent components prevents presentation code from embedding payment calculations or database queries, making each component modular, readily testable, and independently verifiable.

Reason 2: Modifiability and Service Independence

The campus library must interface with varied payment methods (Credit Card, Debit Card, UPI, Campus Wallet) and third-party campus email relays. Isolating external payment handling behind the *IPaymentGateway* interface and database queries behind *IFineData* allows the university to change payment processors, integrate new payment gateways, or alter database schemas without modifying the core Fine Management business logic or student UI.

Security Advantage

The Student Member UI never directly accesses the database or third-party financial gateways. Controlled interfaces create a strict boundary around financial operations. Every payment transaction is mediated by the Business layer, where the Authentication & Session component validates credentials and active login tokens before any financial action proceeds. Confidential payment credentials, API tokens, and database connection strings are confined entirely within server-side components, eliminating client-side tampering and mitigating SQL injection risks.

Performance Benefit

Overdue fine calculations (days overdue × ■5/day, capped at ■200) execute in-memory with sub-millisecond execution times. Database access is strictly indexed on Member ID and Book ISBN for low-latency state retrieval. Furthermore, receipt generation and dispatch are handled asynchronously through the Notification Service, preventing payment confirmation or mail transport latency from blocking the interactive student UI during peak campus operational hours.

Submission Package

This submission contains the complete written architectural justification and the final UML component diagram for Problem Statement #08. The editable Draw.io model source is provided as Digital_Campus_Library_Component_Diagram.drawio, along with high-resolution PNG and SVG exports.